Boosting Sales and Planning Expansion with Data

A Proposal report for the BDM capstone Project

Submitted by

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Contents

Declaration Statement

I am working on a Project titled “Strategic Inventory, Payment, and Pricing Solutions” . I extend my appreciation to **Ramu Enterprises**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: Swati Singh

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Date: 13/11/25

Executive Summary

This project focuses on a small B2B manufacturing company located in Bangalore, engaged in the production of durable and customized doors, UPVC windows, and interior solutions. The organization primarily serves business clients such as builders, contractors, and interior solution providers, operating in a competitive manufacturing and construction-support segment.

The company is currently experiencing a decline in sales due to increasing competition from rival firms. In addition, the organization faces inefficient inventory management, resulting in poor availability of high-demand products and excessive stocking of slow-moving items. This imbalance leads to blocked working capital, increased holding costs, and missed sales opportunities, thereby negatively affecting profitability and customer satisfaction. The owner intends to expand into untapped markets, but lacks data-driven insights to assess demand potential and inventory readiness.

To address these challenges, the project proposes an analysis of historical sales data, purchase data, and inventory records. The solution approach includes demand forecasting techniques to understand demand patterns, ABC inventory classification to prioritize critical and high-value items, and evaluation of inventory optimization practices to reduce stock-outs and excess inventory. The analysis will be carried out using Excel and Python, applying relevant business analytics and statistical methods.

The expected outcome of this project is improved product availability, optimized inventory levels, reduced capital blockage, and enhanced profitability, enabling the organization to respond more effectively to market demand and competitive pressures.

Organization Background

Ramu Doors Private Limited is a for-profit B2B manufacturing company based in Bengaluru, Karnataka, engaged in the production of UPVC doors, flush doors, and premium solid doors for residential, commercial, and industrial applications. The company was founded in 2005 as Ramu Enterprises and has since grown steadily by providing customized door and interior solutions to builders, contractors, and interior designers.

The organization operates as a small-to-medium-scale enterprise, with approximately 25–40 employees across manufacturing, sales, and administrative functions. It follows standard business operations for six days a week and maintains in-house facilities for fabrication, finishing, and installation. In recent years, the company transitioned into a Private Limited entity, reflecting its intent to formalize operations and support future expansion.

Problem Statement

- The company is experiencing a decline in sales as a result of increasing competition from rival firms.

- To address the decline in sales, the owner plans to expand the business into untapped markets and requires data-driven insights and a strategic plan to evaluate the feasibility and potential.

- The company struggles with poor product availability due to inefficient inventory management. The owner seeks a way to optimize inventory to ensure popular products are always in stock.

Background of the Problem

The company is facing multiple challenges that are affecting its growth and market performance. Sales have been declining due to increasing competition from rival firms, resulting in reduced market share and pressure on revenue. To address this issue, the owner is considering expanding the business into untapped markets. However, such an expansion requires careful, data-driven planning to assess market potential, customer demand, and the feasibility of entering new regions effectively. Without proper insights, the risk of unsuccessful expansion and wasted resources remains high.

In parallel, the company is experiencing inefficiencies in inventory management. Popular products are often unavailable because inventory space is occupied by slow-moving or low-demand items. This leads to lost sales, frustrated customers, and missed opportunities to capitalize on high-demand products. Ensuring better product availability through optimized inventory allocation is therefore critical to maintaining customer satisfaction and supporting growth initiatives.

These challenges highlight the need for a comprehensive approach that leverages sales and purchase data to drive informed decisions. By understanding market trends, identifying high-potential areas for expansion, and optimizing inventory management, the company can improve operational efficiency, enhance product availability, and regain a competitive edge in the market.

Problem Solving Approach

This section describes the data-driven and quantitative approaches used to address the business problems identified in Section 3, namely declining sales due to competition and inefficient inventory management leading to poor product availability. The proposed methods focus on structured data analysis to support informed business decisions.

Data Collection and Sources

The analysis will use internal operational data collected from the past 12–24 months. The key datasets include sales data (product-wise sales volume, revenue, order dates, and customer type), inventory data (opening and closing stock levels, stock movement, and stock-out instances), and purchase data (supplier details, purchase quantities, costs, and lead times). These datasets will be sourced from billing records, inventory registers, and purchase logs maintained by the organization.

Tools Used for Analysis

Microsoft Excel and Python will be used for analysis. Excel will support data cleaning, pivot tables, and dashboards, while Python will be used for statistical analysis, regression, and time-series forecasting. These tools are suitable for handling business data and presenting insights in an interpretable manner.

Approach for Addressing Declining Sales

To analyze declining sales and support market expansion decisions, demand forecasting techniques will be applied to historical sales data. Quantitative methods such as time-series analysis and regression models will be used to identify demand trends and seasonality for key products. This analysis will help estimate future demand and align production and inventory planning with market requirements.

Approach for Inventory Optimization

To improve product availability and reduce excess inventory, ABC inventory classification will be conducted based on annual consumption value. This will help prioritize high-value and fast-moving items. Additionally, inventory turnover analysis and reorder points with safety stock calculations will be applied to determine optimal replenishment levels and minimize stock-outs and holding costs.

Monitoring and Evaluation

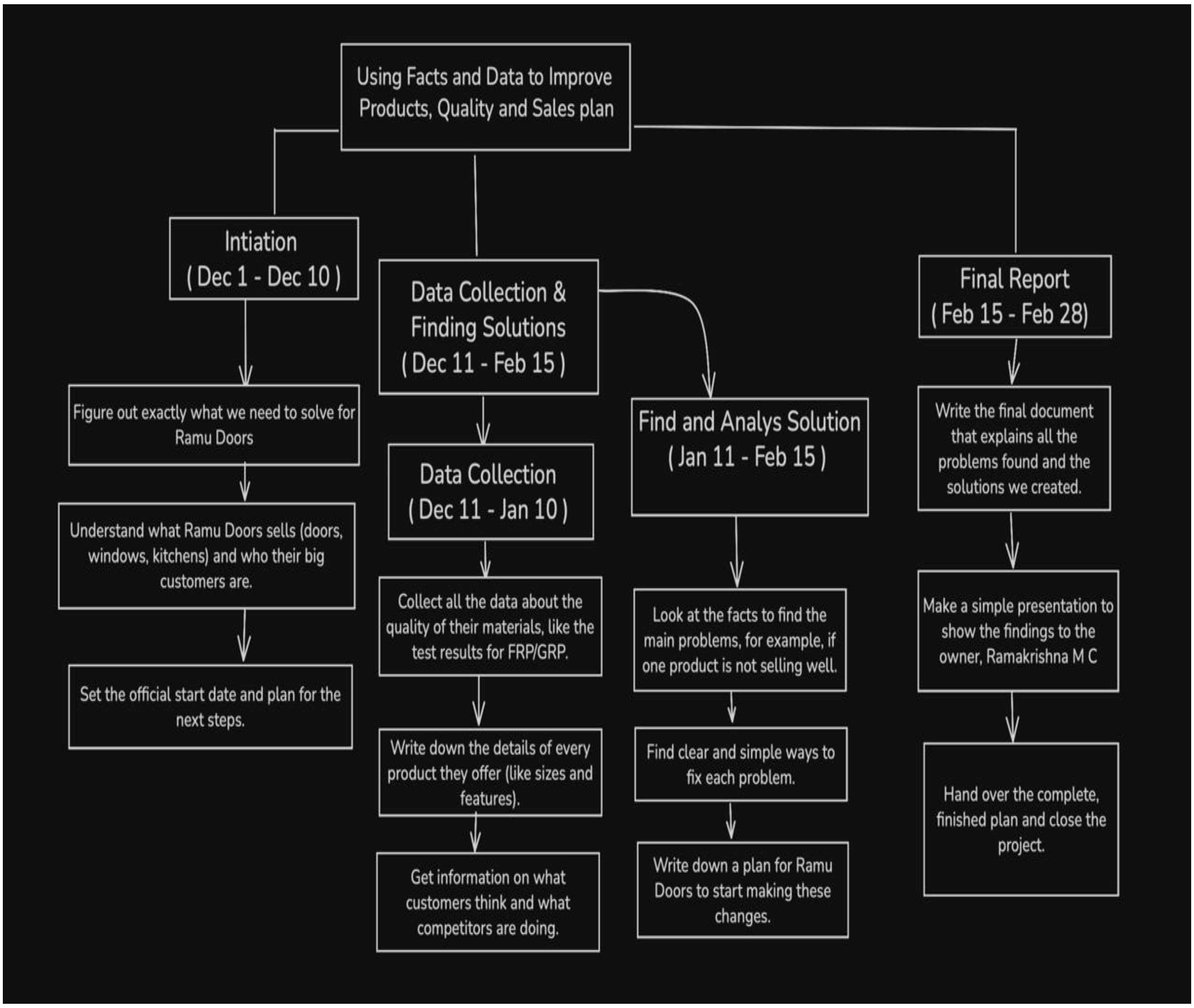
Key performance indicators such as inventory turnover ratio, stock-out frequency, and forecast accuracy will be monitored regularly to evaluate the effectiveness of the proposed solutions and support continuous improvement.

Expected Timeline

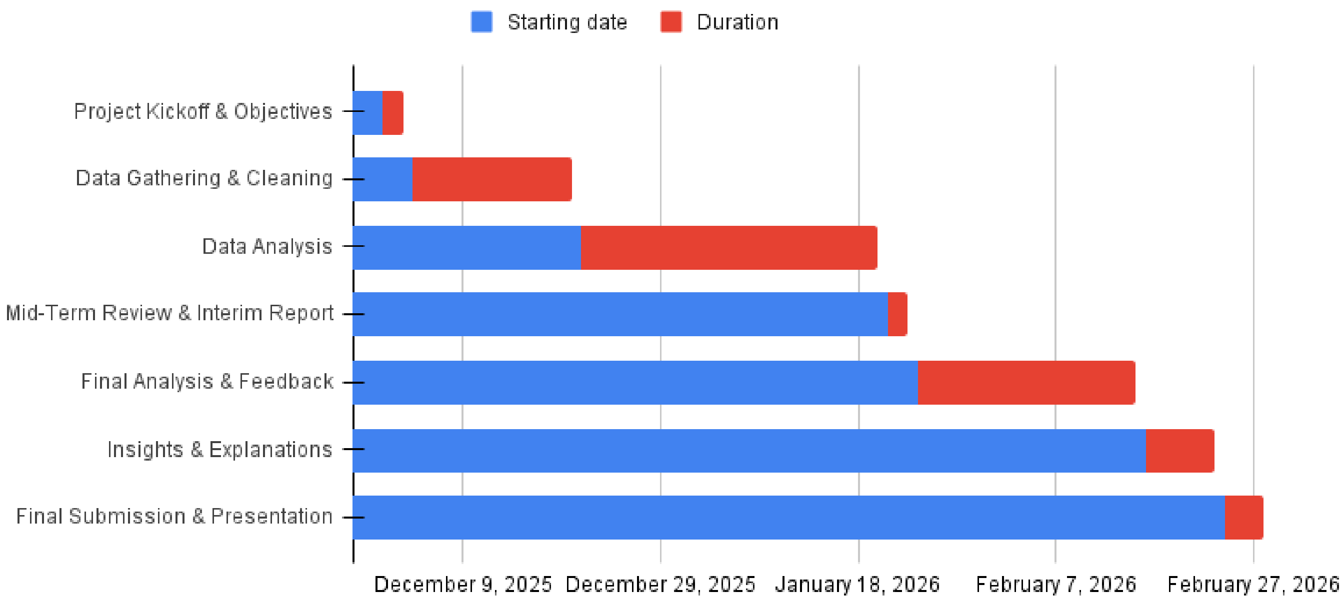
The project timeline for Ramu Enterprises is organized into four clear segments: 1) Initiation, 2) Collection, 3) Analysis, and 4) Reporting.

Each segment outlines a focused set of activities, ensuring that we progress step-by-step from setup to final recommendations, covering the period from November 1, 2025 to January 30, 2026.

Work Breakdown Structure



<u>Gantt chart</u>



Expected Outcome

The expected outcome for Ramu Enterprises, after executing this comprehensive analysis, is a stronger grasp on inventory performance and customer payment patterns, leading to more effective operations overall. By systematically studying which products sell quickly and which accumulate as unsold stock, the company can fine-tune its purchasing and stocking practices, making sure capital is invested in high-demand door SKUs while minimizing losses from slow-moving items. Understanding payment behavior will help flag risky customers in advance and steer credit decisions, thus protecting cash flow. Insights from competitor pricing will position the business to better adjust its own prices, remain competitive, and attract valuable new customers. Ultimately, these results will make inventory turnover healthier, support faster, smarter decision-making, and equip Ramu Enterprises to respond flexibly to market shifts. The business will be better positioned to channel resources where they bring the most value, ensuring long-term growth and resilience.